

The Water Cycle

Name: _____

Date: _____

Year 4 Science — Earth and Space Sciences

ACARA v9.0: AC9S4U02

Learning Intention

I am learning to describe the key processes in the water cycle, including evaporation, condensation and precipitation.

Success Criteria

- I can name and order the stages of the water cycle.
- I can describe what happens at each stage.
- I can explain the water cycle using a real-world example.

4 Key Words: E-C-P-C

Evaporation (water rises as vapour), Condensation (vapour cools, forms clouds), Precipitation (rain/hail falls), Collection (water gathers) — then it repeats!

Worked Example

Where does rain come from?

The sun heats ocean water, which evaporates into vapour, cools and condenses into clouds, then falls as precipitation (rain).

Part A — Warm Up

Circle the stage where water turns into vapour: **evaporation** **precipitation** **collection**

True or False: Water in the water cycle eventually runs out. _____

Name one place water can be collected after it rains. _____

Part B — Core Skills

Order these water cycle stages 1–4:

___ Precipitation ___ Evaporation

___ Collection ___ Condensation

Explain what happens during condensation.

Why does water vapour rise into the sky during evaporation?

Name 2 forms precipitation can take other than rain.

1. _____ 2. _____

Word Bank

evaporation • condensation • precipitation • collection • water cycle • water vapour • cloud • runoff

Working Space — Draw and Label the 4 Stages of the Water Cycle

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Part C — Challenge: The Dam Investigation

A farming town near Geelong relies on a large dam for its water supply. During a hot, dry summer, the dam's water level drops noticeably, but after a week of heavy rain, the level rises again.

Q8. Explain, using water cycle terms, why the dam's water level dropped during the hot summer.

Q9. Explain, using water cycle terms, why the dam's water level rose again after the rain.

Q10. A student says: "Once water evaporates from the dam, it is gone forever." Do you agree or disagree? Use your understanding of the water cycle to explain.

Answer Key — Part A

A1: evaporation

A2: False — the water cycle is continuous; the same water is reused again and again.

A3: A river, lake, ocean, dam or the ground (accept any reasonable answer).

Answer Key — Part B

B1: 2 Precipitation, 1 Evaporation, 4 Collection, 3 Condensation.

B2: Water vapour in the air cools down and changes back into tiny water droplets, forming clouds.

B3: Heat from the sun warms the water, giving it enough energy to change from liquid into water vapour (gas), which is lighter than the surrounding air and rises.

B4: Snow and hail (accept sleet).

Answer Key — Part C

C8: Heat from the sun caused more water to evaporate from the dam than usual, turning it into water vapour and leaving less liquid water behind, lowering the water level.

C9: The rain was precipitation falling from clouds (formed by condensation), and this water collected in the dam, raising its water level again.

C10: Disagree — the evaporated water does not disappear, it rises as vapour, later condenses into clouds, and eventually falls again as precipitation, possibly returning to the same dam or elsewhere.

I Can Checklist

☐ I can name and order the water cycle stages.

☐ I can apply the water cycle to a real example.

☐ I can describe what happens at each stage.